



Experiment-5

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1. Aim/Overview of the practical:

To implement and understand Dynamic Routing using RIP (Routing Information Protocol) and verify communication between different networks using Cisco Packet Tracer.

2. Software/Tools Required:

- Cisco Packet Tracer
- Windows Operating System

3. Theory:

A) What is Routing?

Routing is the process of selecting the best path for forwarding data packets from a source network to a destination network. When devices belong to different networks, routers use routing tables to determine the appropriate path for transmitting data.

B) What is Dynamic Routing?

Dynamic Routing is a routing technique in which routers **automatically learn and update routing information**. Routers exchange routing information with neighboring routers, eliminating the need to manually configure every route.

Advantages of Dynamic Routing

Advantage	Explanation
Automatic Learning	Learns routes automatically
Less Manual Work	No need to configure every route
Fault Tolerance	Can find alternate routes
Scalable	Suitable for large networks

Dynamic Routing Protocols

Protocol	Full Form
RIP	Routing Information Protocol
OSPF	Open Shortest Path First
EIGRP	Enhanced Interior Gateway Routing Protocol
BGP	Border Gateway Protocol

C) RIP (Routing Information Protocol)

RIP (Routing Information Protocol) is a distance-vector routing protocol that automatically exchanges routing information between routers. It is one of the oldest and simplest dynamic routing protocols.

Why was RIP Developed?

RIP was introduced because static routing becomes difficult to manage in larger networks. It helps routers to:

1. Automatically discover networks.
2. Automatically update routing tables.
3. Automatically select routes.

How Does RIP Work?

RIP uses Hop Count as its routing metric to determine the best path. A hop represents a router crossed by a packet while reaching the destination network.

For example:

R1 → R2 → R3 → Network A

From R1, the hop count is 2, because the packet crosses R2 and R3.

RIP Hop Count

Hop Count	Meaning
1	One Router
2	Two Routers
5	Five Routers
15	Maximum Reachable
16	Unreachable Network

RIP Routing Table

Destination Network	Next Hop	Hop Count
192.168.1.0	Directly Connected	0
192.168.2.0	10.0.0.2	1
192.168.3.0	10.0.0.2	2

Feature	Static Routing	Dynamic Routing
Configuration	Routes are configured manually by the administrator.	Routes are learned and updated automatically using routing protocols.
Route Updates	Manual updates are required whenever the network changes.	Routes are updated automatically when the network topology changes.
Scalability	Suitable for small networks.	Suitable for medium and large networks.
CPU Usage	Low, as no route calculations are performed.	Higher, due to continuous route calculations.
Memory Usage	Low memory requirement.	Higher memory requirement for maintaining routing tables.
Bandwidth Usage	Low, since no routing updates are exchanged.	Higher, because routing updates are periodically exchanged.
Security	More secure as routing information is not advertised.	Less secure due to routing information exchange.
Maintenance	Difficult to maintain in large networks.	Easier to maintain in large and changing networks.
Network Changes	Requires manual reconfiguration.	Automatically adapts to network changes.
Best Suited For	Small and stable networks.	Medium and large networks with frequent topology changes.

4. Steps for Experiment/Practical:

A) Dynamic Routing using 6 PCs, 3 Routers and 3 Switches

Procedure / Steps:

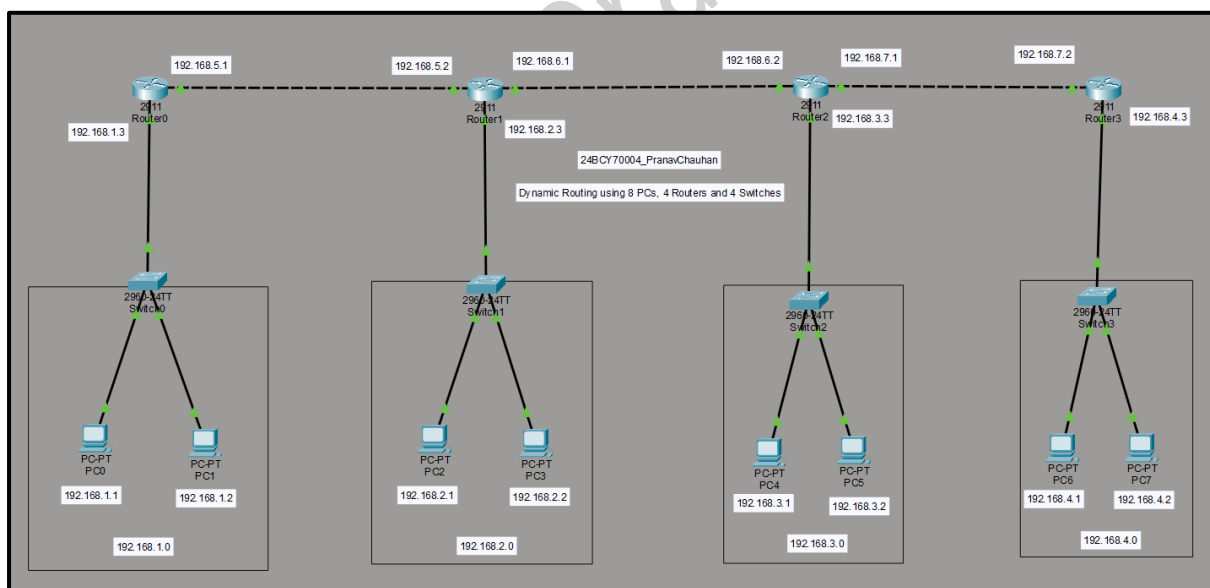
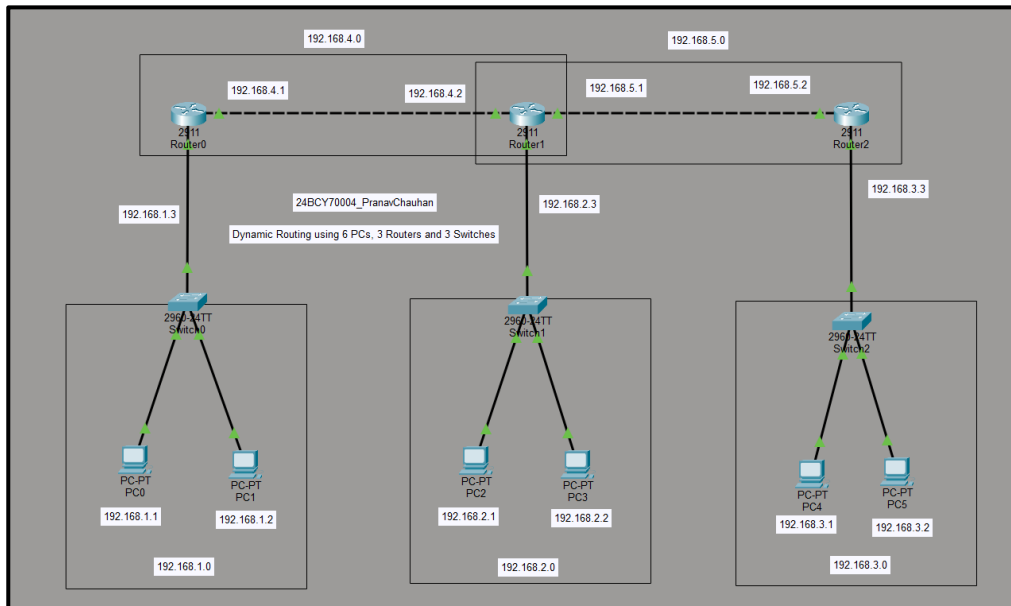
1. Open **Cisco Packet Tracer** and place **6 PCs, 3 Routers, and 3 Switches** in the workspace.
2. Connect all devices using appropriate cables according to the network topology.
3. Assign **IP addresses** to all PCs and configure IP addresses on the router interfaces.
4. Configure the **Default Gateway** on each PC.
5. Configure **RIP** on all three routers and advertise their directly connected networks.
6. Verify the routing table using the command:
`show ip route`
7. Test connectivity between PCs on different networks using the **ping** command.

B) Dynamic Routing using 8 PCs, 4 Routers and 4 Switches

Procedure / Steps:

1. Open **Cisco Packet Tracer** and place **8 PCs, 4 Routers, and 4 Switches** in the workspace.
2. Connect all devices using appropriate cables according to the network topology.
3. Assign **IP addresses** to all PCs and configure IP addresses on the router interfaces.
4. Configure the **Default Gateway** on each PC.
5. Configure **RIP** on all four routers and advertise their directly connected networks.
6. Verify the routing table using the command:
`show ip route`
7. Test connectivity between PCs on different networks using the **ping** command.

5. Output:



6. Result

The implementation of **Dynamic Routing using RIP (Routing Information Protocol)** was studied successfully using Cisco Packet Tracer. RIP was configured on the routers, routing tables were updated automatically, and connectivity between different networks was verified successfully using the **ping** command.

7. Learning Outcomes

- Understood the concept and importance of Routing in computer networks.
- Learned the difference between Static Routing and Dynamic Routing.
- Understood the working of RIP (Routing Information Protocol).
- Learned how RIP uses Hop Count to select the best path.
- Configured RIP on multiple routers and understood automatic route learning and updates.
- Learned to verify routing information using the show ip route command.
- Verified communication between different networks using the ping command.